

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 8: Data Link Protocols (HDLC & PPP)

Course: Computer Networks (Credit Hours: 3)

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1. WHAT ARE DATA LINK LAYER PROTOCOLS?

Data Link Layer protocols handle direct device-to-device data delivery over a single physical cable or connection link. They are responsible for framing, addressing, error detection, and maintaining link synchronization.

The course outline highlights two primary industry-standard protocols at this layer: **HDLC (High-Level Data Link Control)** and **PPP (Point-to-Point Protocol)**.

2. HDLC (HIGH-LEVEL DATA LINK CONTROL)

HDLC is a **bit-oriented** protocol developed by ISO for point-to-point and multipoint communication links.

Key Features of HDLC

- **Bit-Oriented:** Treats data as a continuous stream of bits (not characters).
- **Bit Stuffing:** Uses a special flag byte 01111110. If five consecutive 1s appear in data, it automatically inserts a 0 so the flag is not confused with user data.
- Supports both Half-Duplex and Full-Duplex communication modes.

HDLC Three Frame Types

- **I-Frames (Information):** Carry actual user payload data and piggybacked ACKs.
- **S-Frames (Supervisory):** Carry flow control and error control messages (ACK, NACK, Receive Ready).
- **U-Frames (Unnumbered):** Reserved for link management, setup, and disconnection.

3. PPP (POINT-TO-POINT PROTOCOL)

PPP is a **byte-oriented** protocol widely used to connect home routers and modems directly to an Internet Service Provider (ISP) over telephone lines or leased fiber links.

Key Components of PPP:

- **Framing:** Encapsulates network layer packets using a clear flag pattern (uses Byte Stuffing / Escape characters).
- **LCP (Link Control Protocol):** Establishes, tests, configures, and terminates the physical point-to-point data link.
- **NCP (Network Control Protocol):** Allows multiple network layer protocols (IPv4, IPv6) to run simultaneously over the same physical connection.
- **Authentication Support:** Features built-in password authentication mechanisms: **PAP** (plain password) and **CHAP** (secure encrypted handshake).

4. SUMMARY COMPARISON: HDLC VS. PPP

Feature	HDLC	PPP
Data Orientation	Bit-oriented (uses Bit Stuffing)	Byte-oriented (uses Byte Stuffing)
Network Topologies	Point-to-point and Multipoint links	Strictly Point-to-Point links only
User Authentication	No built-in authentication support	Strong built-in security (PAP and CHAP)
Multi-Protocol Support	Limited; mainly proprietary vendor versions	Full support for multiple protocols via NCP
Common Application	Legacy leased line links (Cisco serials)	Broadband internet connections (PPPoE, DSL)

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why does PPP require an authentication protocol like CHAP?

Ans: Because when you connect to an ISP (via DSL or fiber), the provider must verify your username and password before granting Internet access.

Q2: What is the main difference between Bit Stuffing (HDLC) and Byte Stuffing (PPP)?

Ans: HDLC inserts a single '0' bit after every five consecutive '1' bits, whereas PPP inserts an Escape byte (ESC) before any data character that matches a flag byte.